

MFIN7092 Final Project

Cross-Exchange Funding-Rate Arbitrage in Perpetual Futures: Form a Carry Illusion to a Capacity-Constrained, Risk-Calibrated Strategy

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Abstract

I build a cross-exchange funding-rate arbitrage on 20 perpetual coins and six venues for one year. The naive idea—rank coins by funding and trade the extremes—looks like free money, but it loses money (full-PnL Sharpe -2.54) after I count the price of every leg. Many refinements, including a machine-learning model, all fail out of sample. Two delta-neutral strategies can work. **Strategy A** (same coin, two venues) gets an out-of-sample Sharpe of 2.34 and can be used at any size. **Strategy B** (negative cash-and-carry) reaches 7.47 in regime, but it is off 78% of the time and capped at \$130,926; an honest risk pipeline cuts it to a single-year range of $[2.41, 3.81]$. So A is the main, scalable strategy, and B is only a small one.

1 Introduction

Perpetual futures (“perps”) are the main instrument in crypto trading. A perp never expires; instead the longs and the shorts pay each other a regular *funding* rate, which keeps the perp price near the spot price (when the perp is above spot, longs pay shorts, and below spot it is the opposite). Funding on altcoin perps is often large and stays for a long time, so it looks like easy carry. Can I harvest it in a systematic way?

The answer is no, at least not the naive way. My main finding is negative: the simple way to harvest funding does not only do worse, it *loses money* after I count the price risk honestly. So the rest of the report looks for a way to keep the funding income but remove the price risk.

My contributions are four. First, I show that a funding-only PnL makes the Sharpe look more than twenty points too high. Second, I write down the failed ideas in an open way. Third, I build two delta-neutral strategies and split their Sharpe into clear parts. Fourth, I run a real-data risk pipeline that cuts Strategy B’s naive, 7.47 to an honest range and a hard cap. The sections follow this order.

2 Background and Related Literature

2.1 The funding mechanism

Let P_t be the mark price and I_t the spot index. In each interval the exchange sets a funding rate f_t , which is a premium term

plus an interest term,

$$f_t \approx \text{clip}\left(\frac{P_t - I_t}{I_t} + r_t^{\text{int}}, -\bar{f}, +\bar{f}\right), \quad (1)$$

where $\bar{f}$ is the cap. A long position of size N gets the cash flow

$$\pi_t^{\text{long}} = -f_t N, \quad \pi_t^{\text{short}} = +f_t N, \quad (2)$$

so a long collects funding when $f_t < 0$ and a short collects when $f_t > 0$. [1] show that funding equals the cost of carry of a synthetic spot-plus-financing position. The key point is this: Equation (2) is *only* the funding leg. The total PnL of the perp also has a price term and a cost term,

$$\pi_t^{\text{total}} = \underbrace{-f_t N}_{\text{funding}} + \underbrace{N \frac{P_{t+1} - P_t}{P_t}}_{\text{price}} - \underbrace{c_t}_{\text{cost}}. \quad (3)$$

So if a strategy reports only the funding term, the number is not real, unless the price term is already removed in another way.

2.2 Perpetual pricing and the basis

Define the basis $b_t = (P_t - I_t)/I_t$. Two facts are useful. The same-coin same-venue basis (a perp against its own spot) is small and mean-reverts, so a single-venue hedge is clean—this is the base of Strategy B. The same-coin cross-venue basis (one exchange’s perp against another’s) is larger and slower, because to close it I need capital on two venues at the same time [2]—this is the base of Strategy A.

2.3 Carry, dispersion and the cross-section

Three lines of literature shape the design. The *carry* line [3] treats funding as a risk premium that depends on the state. The *arbitrage and dispersion* line [2] shows that cross-exchange gaps mean-revert but are limited by how fast capital can move. The *factor* line [4, 5] prices the cross-section, and I use it as a benchmark. I also use [6] for sizing [7] for cost, [8] and [9] for honest testing, [10] for the ML part, and [11] for crowding.

2.4 Funding as a price of leverage

Funding is the interest of leverage: a long perp is a spot long with leverage, and without frictions funding would be near the risk-free rate [1]. The large and often *negative* funding I find

(Section 4) is the gap from this benchmark, and its size shows how strong the frictions are. Negative funding means traders are willing to pay to be short, that is, the natural longs are too few. So to harvest funding is to give missing this leverage—it is a paid risk, not a free profit.

2.5 How this paper relates to prior work

Most crypto-arbitrage papers study the *price* gap across venues [2]. Strategy A moves this idea to the *funding* basis: I trade the slow and lasting cross-venue funding difference, not the fast price gap. So the contribution of this paper is not a new anomaly, but a careful, friction-aware build and de-risk of two usable strategies, with an honest count of how much edge is left after price risk, borrow limit, and regime.

3 Methodology: Data and Backtest Engine

3.1 Data

Table 1 lists the inputs: an Auros funding feed (about 365,000 events), 65 GB of Tardis tick data put into a 4-hour panel, OKX and Gate.io spot, and live OKX borrow rates and quotas.

Table 1: Data sources and roles.

Input	Source	Role
Funding history	Auros feed	primary signal
Derivative ticker	Tardis (65 GB)	mark / OI / basis
Spot prices	OKX, Gate.io API	cash-and-carry
Borrow / quota	OKX lending API	B financing + cap
Fees	VIP9 schedules	cost model

The universe is 20 coins on six venues (Binance, Bybit, Gate.io, Hyperliquid, KuCoin, OKX), which gives 118 coin-venue pairs. These coins survived six to twelve months, so there is a survivorship bias, and I measure its cost in Section 10.

3.2 The 4-hour decision grid

Funding settles on different clocks (from hourly to 8-hourly), so I use a 4-hour grid that lets every venue fall on a bin edge. To avoid look-ahead, the funding I collect in a bin is set just before the bin ends so the rate I see when I decide is never the rate that pays. The out-of-sample (OOS) period is the last six months (1,086 bins); the first six months are in-sample, for any parameter I must choose.

3.3 The backtest engine

All strategies use one engine. At each bin it computes a signal, chooses positions, and holds them with a *stickiness* rule: a position opens only above an entry threshold θ_{ent} , and closes only below an exit threshold $\theta_{\text{ext}} < \theta_{\text{ent}}$, which is the toxic-arbitrage buffer of [8]. The sizing follows [6],

$$w_{i,t} = \text{clip}\left(\frac{v^*}{\hat{\sigma}_{i,t}}, w_{\min}, w_{\max}\right), \quad (4)$$

with $v^* = 150$ bp, $w_{\min} = 0.3$, $w_{\max} = 1.5$, and $\hat{\sigma}_{i,t}$ lagged by one bin.

3.4 Cost and fill model

Every leg pays the VIP9 maker fee plus 1 bp slippage, and I count the, round-trip cost only when a position changes,

$$c_t = \sum_{i \in \Delta_t} (\text{maker}_{e(i)} + 1 \text{ bp}), \quad (5)$$

where Δ_t is the legs opened or closed at t . Fills are a probability, not a sure thing: a signalled entry is taken with probability 0.8, and the entry-bin price PnL is cut by 12.5%. All Sharpe ratios are after cost and are annualised with $\sqrt{365 \cdot 24/4}$.

3.5 Evaluation philosophy

The main number is the annualised Sharpe,

$$\text{Sh} = \frac{\bar{r}}{\hat{\sigma}_r} \sqrt{N_{\text{yr}}}, \quad N_{\text{yr}} = \frac{365 \cdot 24}{4}. \quad (6)$$

Two choices come again and again. For an on-off strategy (B) I report a bootstrapped *range*, not one point, because the in-regime Sharpe is not what a live book earns. For any regression I use Newey–West errors [12], because funding and basis are autocorrelated.

3.6 Look-ahead controls and validation

Overfitting can come from three doors, and I close each one. *Future information*: every signal uses only data before t (the collected funding, the lagged volatility, and the forward price). *Parameter selection*: I choose parameters on the first six months only, and report all results on the untouched last six months. *Survivorship*: Section 10 adds phantom delistings at a live-measured rate. Because I tried many variants, I treat the winners as upper bounds and report the deflated Sharpe [9].

4 Exploratory Data Analysis

First I look at the funding data, because its shape decides which strategies can work

4.1 The shape of funding

Figure 1 shows a sharp peak at zero and a heavy negative tail, so a few coins carry most of the money—this is where Strategy B lives.

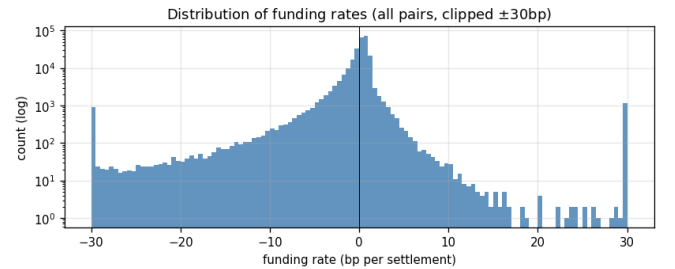


Figure 1: Distribution of funding rates. Pooled across all 118 pairs (clipped to ± 30 bp, log count axis). A sharp zero-peak with a heavy negative tail—the negative tail is where Strategy B lives.

Table 2 splits this by coin. The large caps have small positive funding and no carry; the mid-caps and meme coins (BERA,

KAIITO, IP, MOODENG) stay deeply negative and pay a long a large yearly subsidy. This is the reason for Strategy B.

Table 2: Funding summary statistics by coin (per settlement).

Coin	Mean (bp)	Std (bp)	Neg (%)
BTC	+0.32	0.56	21.6
ETH	+0.28	0.69	27.0
LTC	+0.35	0.86	24.6
DOGE	+0.34	1.30	26.9
SOL	+0.07	1.52	38.8
AVAX	+0.06	1.68	36.7
TAO	+0.10	1.64	25.3
POPCAT	+0.21	1.83	18.6
WLD	+0.11	1.65	31.9
TIA	−0.01	1.68	32.5
JUP	−0.13	1.34	38.1
TRUMP	−0.29	2.23	45.3
JTO	−0.27	5.38	28.0
MOODENG	−1.29	5.72	41.0
IP	−2.01	7.04	53.4
KAIITO	−2.47	10.74	51.3
BERA	−3.42	9.50	75.7
PIPPIN	+19.1	52.3	12.6

4.2 Funding differs across venues

Funding also differs across venues for the same coin, and this is the raw material of Strategy A. In Table 3 OKX is special (mean +3.59 bp, spread 24.6 bp), while KuCoin is negative for more than half the time.

Table 3: Funding summary statistics by exchange.

Exchange	Mean (bp)	Std (bp)	Neg (%)
Binance	−0.60	5.25	36.7
Bybit	−0.19	4.43	26.4
Gate.io	−0.45	5.67	29.4
Hyperliquid	−0.11	1.19	30.6
KuCoin	−0.86	5.96	52.8
OKX	+3.59	24.56	31.9

Figures 2 and 3 show that, coins differ (rows) and venues differ (columns), and that venues move together but are not the same. The left gap closes only slowly [2].

4.3 Persistence: why ML is useless

Funding changes very slowly: the lag-1 (4-hour) autocorrelation is 0.57, and it is still 0.27 one day later (Table 4, Figure 4). So the simple forecast “next = current” is hard to beat, and a model cannot help (Section 6: OOS R^2 0.70 for the simple rule against 0.23 for LightGBM).

4.4 The cross-exchange spread is large and frequent

Strategy A trades the cross-exchange funding spread g . Its median is 8.8 bp and its 99th percentile is 15.9 bp (Table 5),

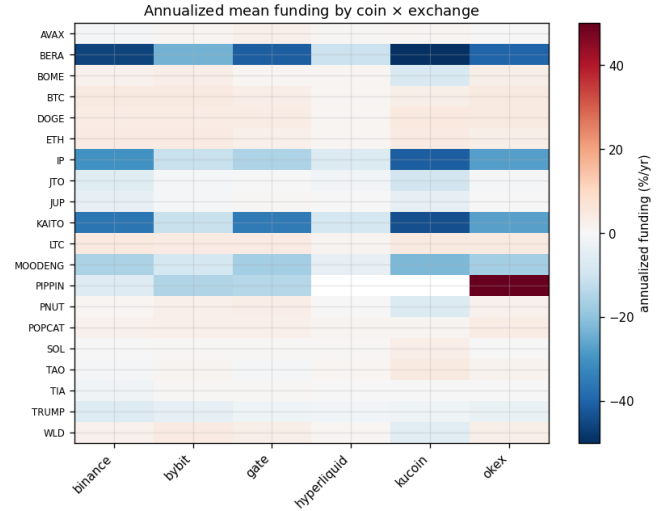


Figure 2: Annualised funding by coin $\times$ exchange. Coins differ (rows) and venues differ (columns); the within-row variation is the spread Strategy A trades.

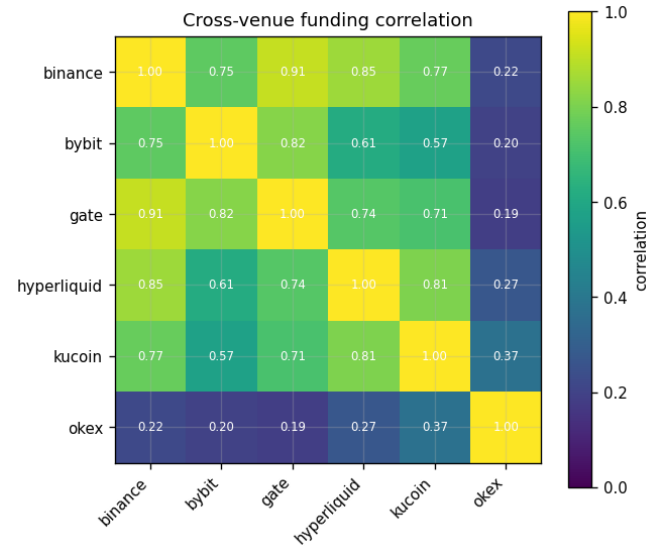


Figure 3: Cross-venue funding correlation. Venues move together but are far from the same; the leftover difference is what a same-coin pair earns.

Table 4: Funding persistence (cross-pair mean ACF).

Lag	Autocorrelation
1 (4h)	0.571
3 (12h)	0.345
6 (24h)	0.268
12 (48h)	0.188
24 (96h)	0.134

Table 5: Cross-exchange funding-spread percentiles (bp).

	p50	p75	p90	p95	p99
Spread	8.8	9.8	10.8	11.9	15.9

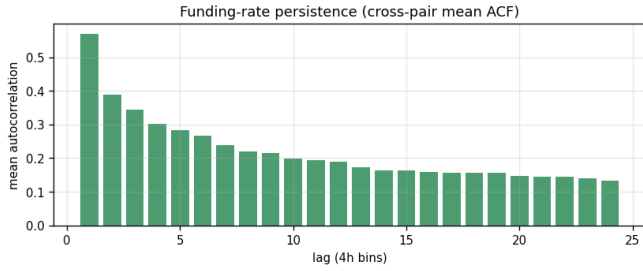


Figure 4: Funding-rate persistence. The autocorrelation starts at 0.57 and falls slowly. This explains why funding is harvestable and why a model cannot beat the simple forecast.

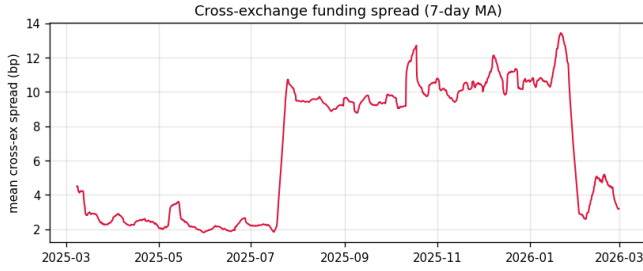


Figure 5: Cross-exchange funding spread over time (7-day MA). The spread stays above Strategy A's 3 bp entry threshold and widens in stressed periods.

both well above the 3 bp entry threshold, so it fires often; it also widens in stress (Figure 5).

Figure 6 confirms the coin choice: Strategy B's five coins is negative most often.

4.5 Funding paths through time

Figure 7 shows that BTC stays near zero, while KAITO and BERA stay deeply negative for long times and then jump back. So the carry is on-off, not constant—this is the root of the regime risk in Section 10.

4.6 Liquidity: open interest

Depth differs by orders of magnitude (Figure 8). This is why Strategy A uses only the eight most liquid coins, and why Strategy B's thin mid-caps make its size only tens of thousands of dollars.

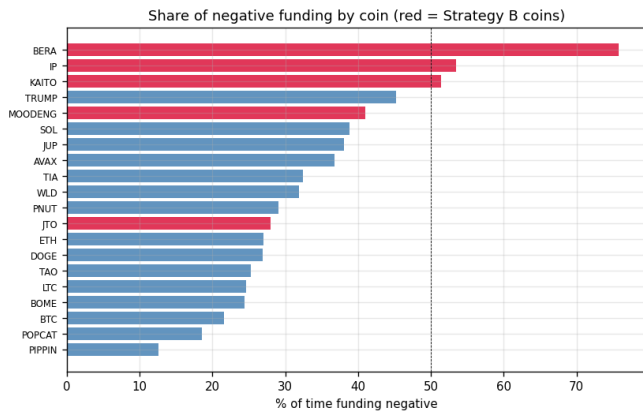


Figure 6: Share of negative funding by coin. Strategy B's five coins (red) are negative far more often than the majors, so the cash-and-carry edge lives in the mid-cap/meme group.

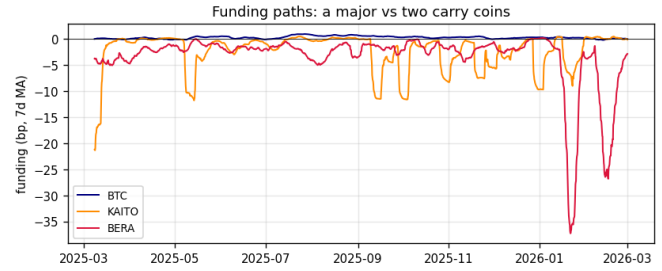


Figure 7: Funding paths: a major versus two carry coins. BTC stays near zero; KAITO and BERA spend long episodes deeply negative. The carry is on-off, the root of the regime risk in Section 10.

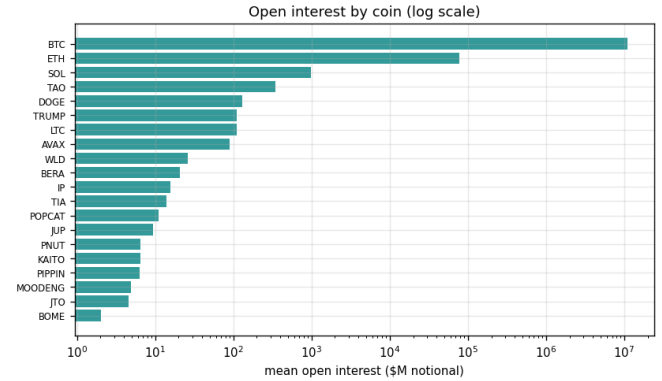


Figure 8: Open interest by coin (log scale). Depth spans several orders of magnitude; this gradient sets which coins each strategy can trade and at what size.

4.7 The basis

Over all pairs, the mark-minus-index basis has mean, -5.3 bp and standard deviation 13.7 bp (Figure 9); this spread makes the basis tradeable, and its mean reversion is the second engine of Strategy A.

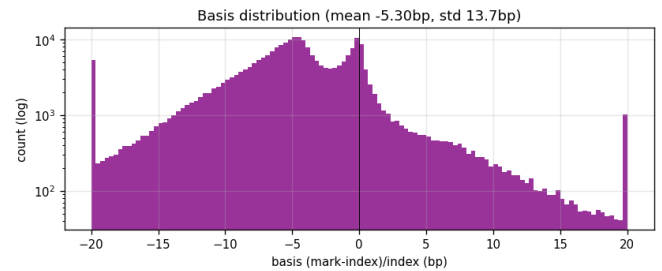


Figure 9: Basis distribution (mark - index). Mean -5.3 bp, standard deviation 13.7 bp, with heavy tails; the spread is the raw material of the basis-convergence engine.

4.8 Intraday seasonality

There is a small hour-of-day pattern (Figure 10), but it is small next to the cross-sectional and regime variation, so I do not use it.

4.9 Mean reversion: what is tradeable

Figure 11 compares two things. The funding spread lasts (lag-1 0.57, half-life about 5 h), so I can catch it on a 4-hour grid. The price basis reverts almost at once (lag-1 -0.04 , half-life under 15 min), so to trade the cross-exchange *price* gap directly is

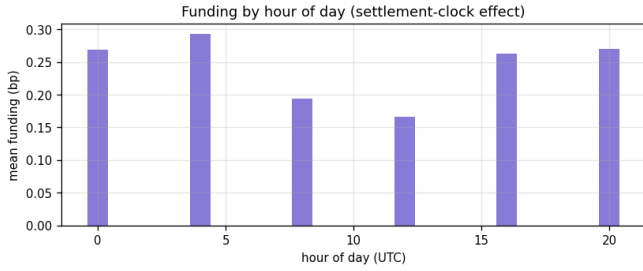


Figure 10: Funding by hour of day (UTC). A mild settlement-clock pattern exists but is small next to the cross-sectional and regime variation the strategies use.

hopeless. So Strategy A trades the lasting funding spread, not the price gap.

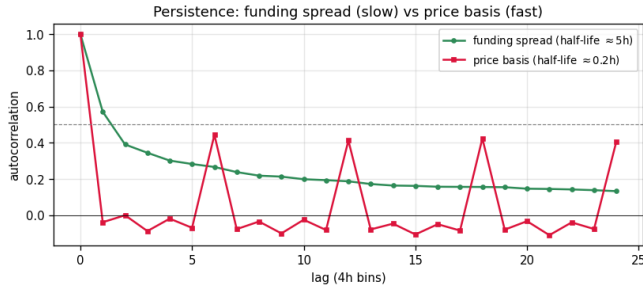


Figure 11: Persistence: funding spread versus price basis. The funding spread reverts slowly (half-life ≈ 5 h, tradeable); the price basis reverts almost at once (half-life < 15 min, untradeable). Strategy A trades the funding spread.

5 The Funding-Carry Illusion: A Naïve Baseline

The first obvious trade ranks all 118 pairs by funding and trades the extremes: long the 24 most-negative, short the 24 most-positive, equal weight, no hedge. On the *full* PnL of Equation (3), it fails (Figure 12): the funding part goes up (+0.38), but the unhedged price part loses a lot (−0.79), so the Sharpe is −2.54 and the return is −90% a year.

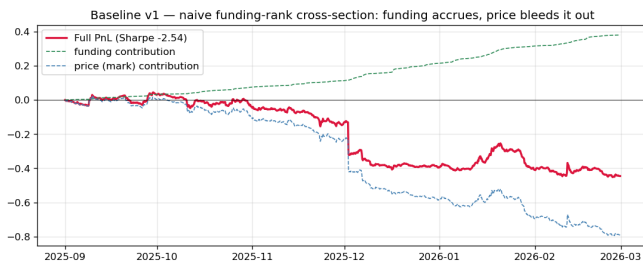


Figure 12: Naïve funding-rank cross-section (full PnL). Funding accrues steadily (green), but the unhedged price term (blue) dominates and the net book (red) loses: Sharpe −2.54.

The reason is simple. The long leg, and the short leg are *different coins*, so the book is not market-neutral; its cross-sectional variance is one to two orders of magnitude larger than the small funding it wants to collect. If I report only the funding term the Sharpe looks above +27, but this is not real. So the rest of the report keeps the funding but removes the price—across venues (A) or against spot (B).

6 A Catalogue of Failed Attempts

Before the strategies that work, I write down the ones that fail, because a process that reports only winners looks the same as overfitting.

6.1 Machine-learning funding prediction

I used LightGBM to predict the next funding [10]. It lost to the simplest forecast: OOS $R^2 = 0.23$ against 0.70 for “next = current”. Funding changes too slowly for a model to add anything, so I dropped it.

6.2 Universe expansion and the look-ahead trap

To add more coins by an open-interest cutoff seemed to help (Sharpe 3.09), but only when I chose the cutoff on the full sample; when I chose it on training the window only, it fell to 1.82. So the gap was pure look-ahead.

6.3 Fifteen microstructure and sizing variants

I tested fifteen more ideas (Table 6): threshold tiers, gap sizing, an AR(1) predictor, toxic-move and crowding filters [8, 11], sign rules, top- K , persistence gates, and volatility swaps. None of them clearly beat the Strategy A baseline out of sample. So the edge is in the structure, not in the add-ons.

Table 6: The cold list: fifteen rejected variants (strict OOS Sharpe).

Variant	Sharpe
Tier-scaled threshold (T1=1.5bp, T2=3bp)	1.60
Position size $\propto \sqrt{\text{gap}}$	1.66
Position size $\propto \text{gap}$ (linear)	1.74
AR(1) basis prediction (enter early)	0.66
Toxic-move filter [8]	1.66
OI z-score > 2 skip [11]	1.82
Require sign reversal	1.53
Top- K cross-section	1.83
Signal persistence ($k = 2$ bins)	0.98
EWMA vol replacement (24h)	1.93
Rolling-48h vol replacement	1.87
Regime-adaptive threshold	1.60
BTC delta hedge (already neutral)	2.34
Positive cash-and-carry	−1.15
Multi-stream tangency	2.40

6.4 Positive cash-and-carry

The opposite positive cash-and-carry had an OOS Sharpe of −1.15: in this, sample only the negative-funding side paid, and this side became Strategy B. Figure 13 shows that none of the rejected curves pulls away.

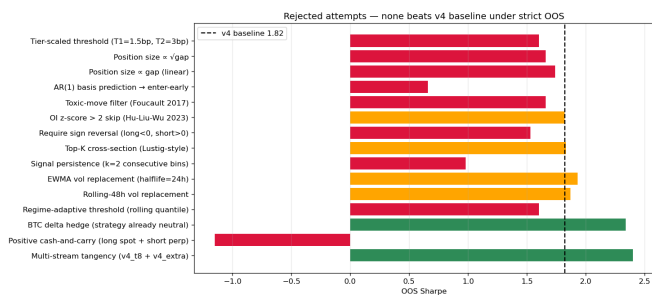


Figure 13: Rejected variants. Equity curves of the main failed ideas. None gave a robust, scalable gain over the simple delta-neutral pair of Section 7.

7 Strategy A: Cross-Exchange Funding-Spread Capture

7.1 Construction

Strategy A keeps the funding income but removes the price, by trading the *same coin* on two venues. For each coin I watch the cross-exchange spread $g = \max_e f_e - \min_e f_e$. When $g \geq 3$ bp I opn a delta-neutral pair: I long the perp on the lowest-funding venue (it pays me as a long) and short the perp on the highest-funding venue (it pays me as a short). Because both legs are the same coin, the price cancels, and what is left is the funding difference plus a small cross-exchange basis that mean-reverts [2]. I hold until the spread turns over or a 1.5 bp exit [8], size by Equation (4), and use only the eight most liquid coins.

7.2 Performance

In the six-month OOS window, Strategy A gets a Sharpe of **2.34**, 8.8% a year, 3.7% volatility, and a max drawdown of only -0.9% (Table 7, Figure 14)—a stable relative-value profile, the opposite of the baseline.

Table 7: Headline performance (OOS).

Strategy	Sharpe	Ann. Ret	Vol	MD
Baseline v1 (naïve)	-2.54	-90%	—	—
Strategy A (v4)	+2.34	+8.8%	3.7%	-0.9
Strategy B (in-regime)	+7.47	+190%	25.5%	-5.0

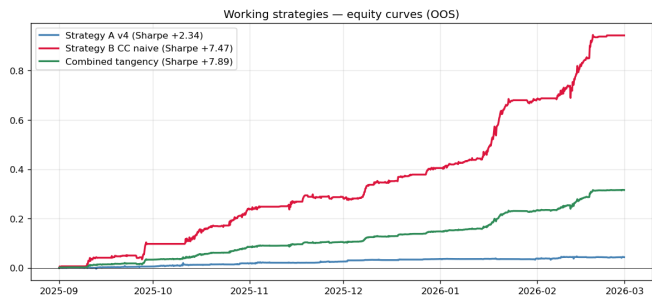


Figure 14: Strategy A equity (OOS). The delta-neutral cross-exchange pair grows smoothly to a Sharpe of 2.34 with a maximum drawdown of -0.9% .

7.3 The two engines

Figure 15 splits the PnL into funding and basis: the two parts are about the same size and almost independent, so when one engine is quiet, the other one often is not.

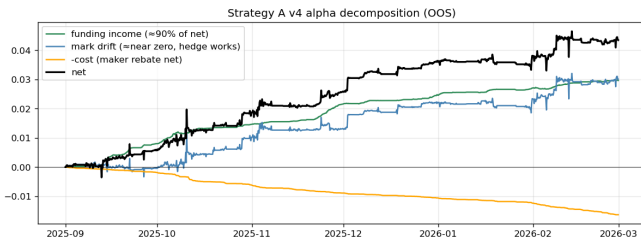


Figure 15: Strategy A: funding versus cross-exchange basis. The two parts are about the same size and almost independent—two engines, not one.

7.4 Sizing

Volatility-target sizing adds about half a Sharpe point (Figure 16), because it sizes up when the pair is calm and down when it is wild.

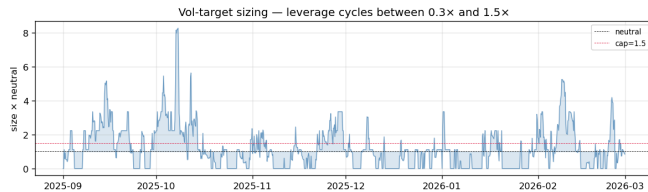


Figure 16: Volatility-target sizing. Scaling against volatility (target 150 bp, cap $1.5\times$) adds about $+0.5$ Sharpe by cutting exposure when pair volatility spikes.

7.5 Stability and return profile

Three more checks support the stability: the 30-day rolling Sharpe stays positive (Figure 17), the worst drawdown is -0.9% with fast recovery (Figure 18), and most months are positive (Figure 19).



Figure 17: Strategy A: 30-day rolling Sharpe. The rolling value stays positive and moves around the full-sample 2.34 (dashed): the edge is steady, not a one-off.

The returns are not Gaussian: the sekw is $+1.83$ and the excess kurtosis is near 40 (Figure 20), so the Sharpe makes this shape look too good, and I treat the tail in Section 10.

7.6 Activity and breadth

The book holds and rotates only a few pairs (Figure 21); it is neither a monthly rebalance nor a fast churner.

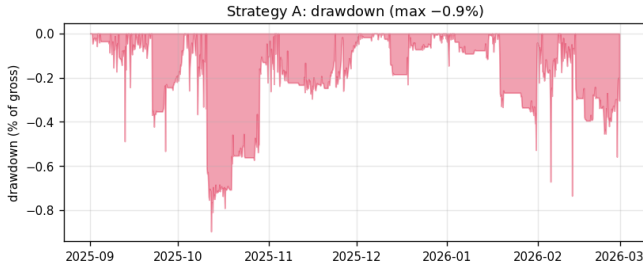


Figure 18: Strategy A: drawdown. The maximum drawdown is -0.9% with fast recoveries—the sign of a relative-value trade.



Figure 19: Strategy A: monthly returns (OOS). Positive in most months and not driven by a single outlier.

7.7 Attribution by coin

The PnL is broad: six of the eight coins are positive (Table 8, Figure 22), and this is why A can scale but B cannot.

7.8 Parameter sensitivity

When I sweep the two free parameters (Tables 9–10, Figures 23–24), the 3 bp threshold is at the top of a smooth single peak, and the volatility target has a wide flat region (100–200 bp). The joint heatmap (Figure 25) shows one connected high-Sharpe area, not a single spike, so this is not overfitting.

8 Strategy B: Negative Cash-and-Carry

8.1 Construction

Strategy B removes the price inside a *single* venue. On OKX, when funding is deeply negative ($f \leq -8$ bp), I long the perp (collecting $-f$) and short the spot of the same coin, and I fund the short by borrowing the coin. The delta cancels, and what is left is the funding minus the borrow cost; I exit when funding

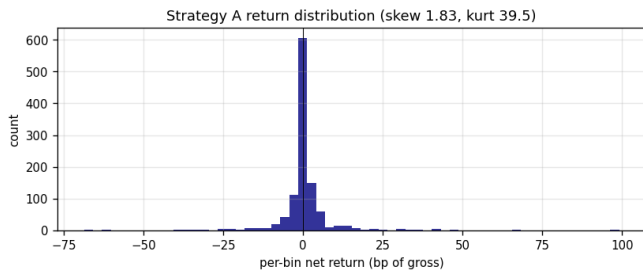


Figure 20: Strategy A: per-bin return distribution. Positive skew ($+1.83$) and high kurtosis (≈ 40)—the carry shape. The Sharpe shows the centre but understates the tail.

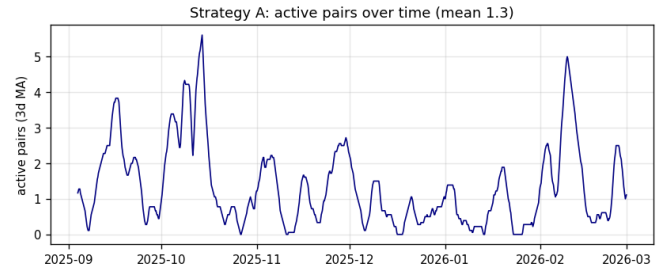


Figure 21: Strategy A: active pairs over time. The book holds and rotates a handful of pairs, spread enough that no single gap dominates.

Table 8: Strategy A: PnL share by coin.

Coin	Share	Coin	Share
TRUMP	36.8%	SOL	5.1%
TAO	30.8%	LTC	4.5%
AVAX	14.9%	BTC	1.9%
ETH	6.2%	DOGE	-0.3%

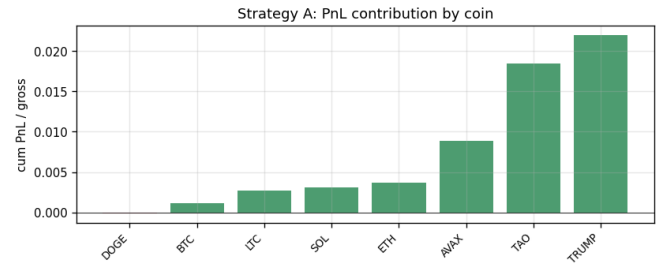


Figure 22: Strategy A: PnL contribution by coin. Six of eight coins are positive; the edge is broad, not concentrated.

Table 9: Strategy A Sharpe versus entry threshold.

Threshold (bp)	1	2	3	4	5	6	8
Sharpe	0.21	1.27	2.34	1.08	0.14	-0.62	-0.52

Table 10: Strategy A Sharpe versus volatility target.

Vol target (bp)	50	100	150	200	300	off
Sharpe	0.59	1.86	2.34	2.33	2.24	1.82

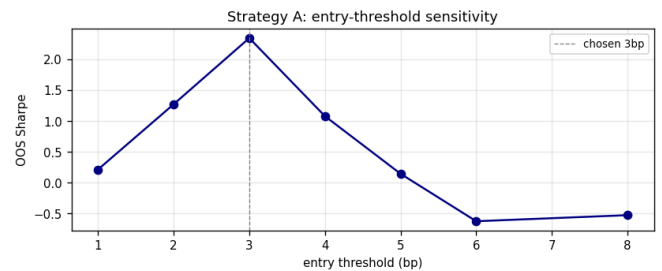


Figure 23: Entry-threshold sensitivity. A smooth, single-peak curve with the top at the chosen 3 bp.

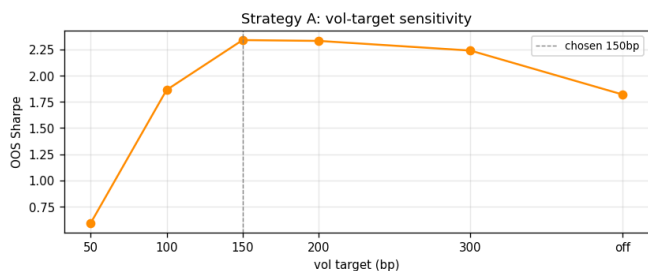


Figure 24: Volatility-target sensitivity. A wide flat region from 100–200 bp; the chosen 150 bp is near the top and “off” falls to 1.82.

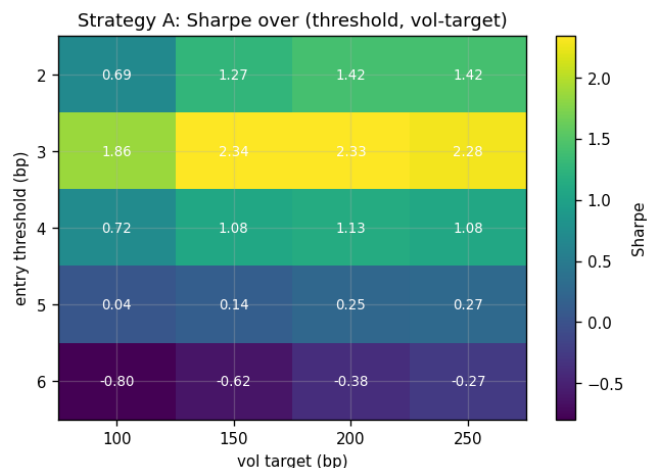


Figure 25: Joint parameter sensitivity. Sharpe over the (threshold, vol-target) grid: one connected high-Sharpe area, not a lone spike—evidence against overfitting.

risers above -2.4 bp. Because the hedge is against the coin’s own spot, the left basis is much smaller than A’s cross-venue one, so B is almost a pure funding play.

The borrow leg is both the cost and the limit. Borrow rates change over time (so I use the real path), they rise when funding is most negative (I stress-test up to $5\times$), and they have a quota (the pool can run out). The first two are survivable; the third one is the real limit (Section 10).

8.2 Performance and concentration

On the surface B looks very strong: in-regime Sharpe 7.47, 190% a year, and a max drawdown of -5.0% . But two, facts decide the rest. The PnL is 99% in five coins (Figure 26), and the trade is off for most of the time and limited by borrow. Such a big number on five meme coins make should me careful, and Section 10 is this careful check.

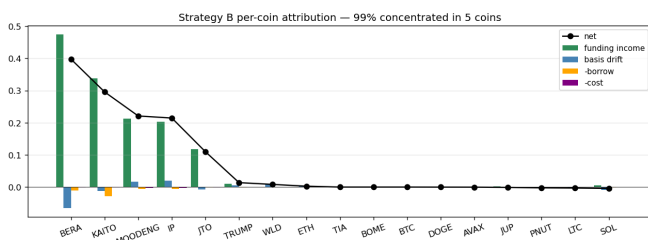


Figure 26: Strategy B attribution. Ninety-nine percent of the cash-and-carry PnL comes from five mid-cap/meme coins with funding that stays deeply negative—the same coins that carry the blow-up and borrow risk.

Figure 27 fast shows in-regime compounding with mild drawdowns—the nice picture that the risk pipeline cuts down.

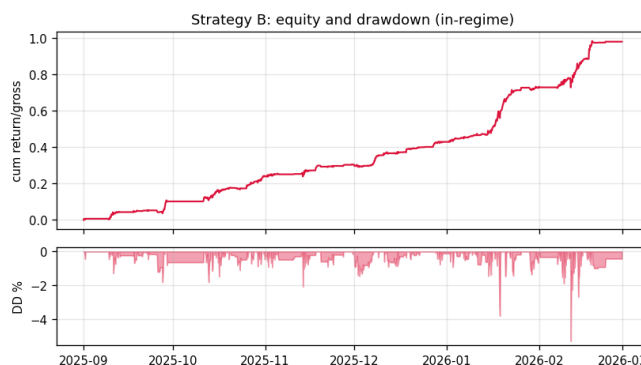


Figure 27: Strategy B: in-regime equity and drawdown. Fast compounding with mild drawdowns—the tempting picture the risk pipeline of Section 10 is built to cut down.

Across the five coins the PnL is fairly even (Table 11, Figure 28); the concentration just follows where the negative funding is.

Table 11: Strategy B: PnL share by coin.

Coin	Share	Coin	Share
BERA	29.3%	MOODENG	15.6%
KAITO	25.6%	JTO	9.7%
IP	20.1%		

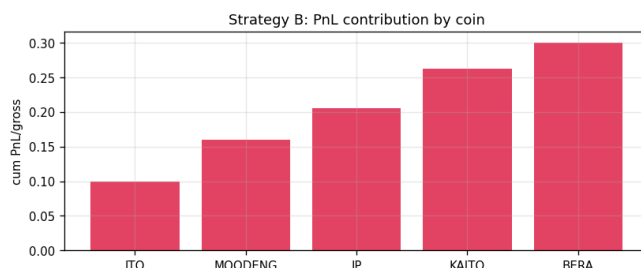


Figure 28: Strategy B: PnL contribution by coin. Held to five names but fairly even among them; the concentration follows where negative funding lives.

9 Where Does the Sharpe Come From?

I split each Sharpe into cler, uncorrelated parts (Table 12, Figure 29). Strategy A is **dual-source**: funding (+1.60) and cross-exchange basis (+1.62) add about the same and are independent (correlation -0.06). Strategy B is **single-source**: it is funding (+8.13) only, and the same-exchange basis is near zero, so the hedge is clean. A’s basis (cross-venue) and B’s basis (perp against spot) are not the same thing, and I should not mix them.

10 Risk Analysis

Strategy B’s naive 7.47 should make me careful. I cut it down step by step, using real data as much as I can. A needs less of this, so the pipeline below is for B.

Alpha source decomposition — A vs B, separately

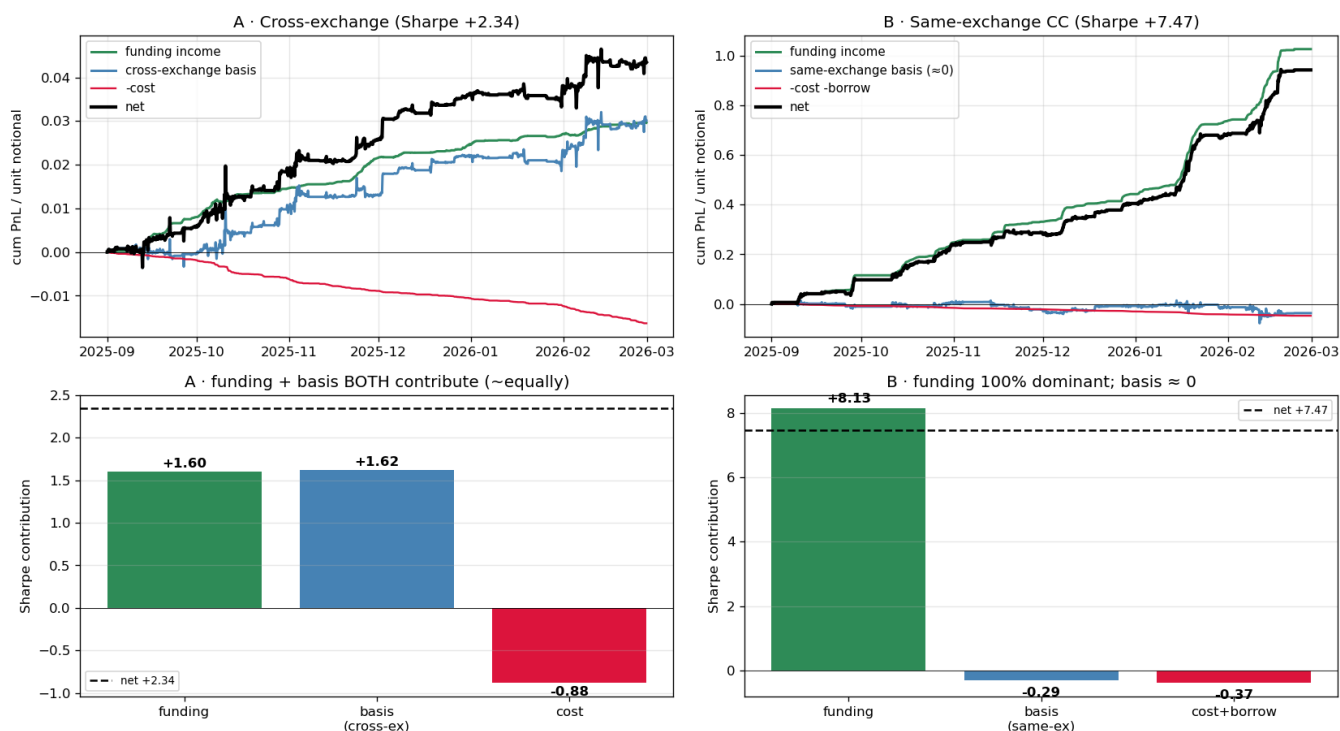


Figure 29: Alpha decomposition, A versus B. Strategy A has two independent engines (funding spread and cross-exchange basis convergence); Strategy B is a pure single-venue funding carry. The mechanisms differ, which is why the two strategies are almost uncorrelated.

Basis drift per 4h — already in strategy PnL (perp + spot legs)

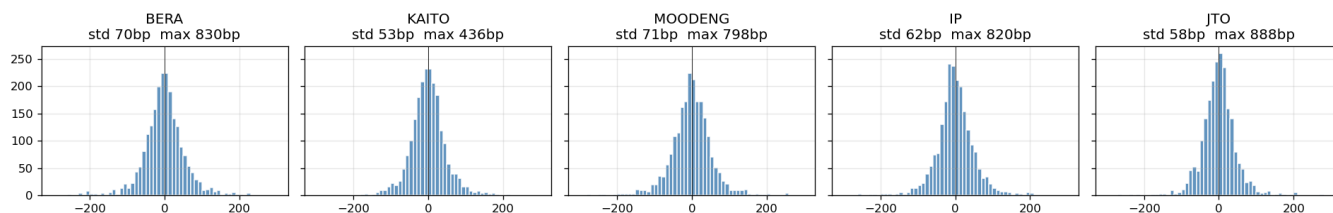


Figure 30: Inventory / basis-drift risk. Per-coin basis drift of 53–71 bp per 4-hour bin is booked into the PnL, so the leftover price risk is captured, not ignored.

Table 12: Sharpe decomposition by source (OOS).

Source	Strategy A	Strategy B
Funding income	+1.60	+8.13
Cross-exchange basis	+1.62	—
Same-exchange basis	—	-0.29
Cost (+ borrow)	-0.88	-0.37
Net Sharpe	+2.34	+7.47
Corr(funding, basis)	-0.06	+0.00

10.1 Is inventory risk actually in the PnL?

The price risk of the hedge is in the PnL: the basis moves are, counted in every bin, with a drift of 53–71 bp peer 4 hours (Figure 30). So the hedge is real, not assumed.

10.2 Survivorship and phantom delistings

The universe survived, so I add *phantom delistings*. At industry rates, 5–10 in a year bring the Sharpe to 5.5–6.5 (Figure 31). A live snapshot supports this: 16.5% of Binance perps were

being delisted (Figure 32). About one fifth of these coins is removed at any time, so a live book must assume that its best coins can disappear.

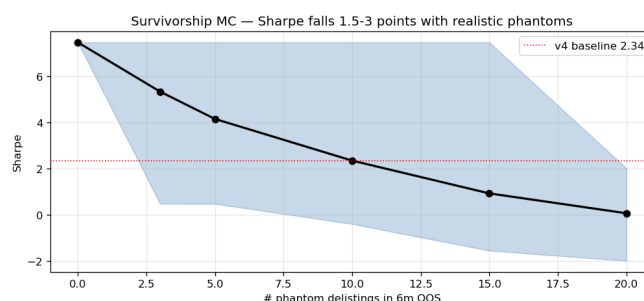


Figure 31: Phantom-delisting Monte-Carlo. Adding 5–10 industry-calibrated phantom delistings over a year cuts the in-regime Sharpe from 7.5 to 5.5–6.5.

10.3 Asymmetric halts and coin blow-ups

Mid-cap coins jump (IP -53%, PIPPIN -63% in a day, Table 13). A 20%/yr halt rate, brings the Sharpe to about 5, and

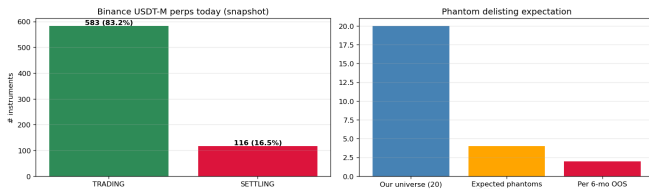


Figure 32: Real-time survivorship. A live snapshot shows 16.5% of Binance perps being delisted, so the haircut rests on measurement, not assumption.

a 50%/yr crash to about 3.5 (Figure 33). These jumps are one-sided, so I usse only hlaf of the empirical rate.

Table 13: Empirical coin blow-up statistics.

Coin	Worst 24h	P(< -30%)
BERA	-37.4%	0.26%
KAITO	-27.5%	0.00%
MOODENG	-32.6%	0.03%
IP	-53.4%	0.39%
JTO	-38.2%	0.15%
PIPPIN	-63.0%	1.25%

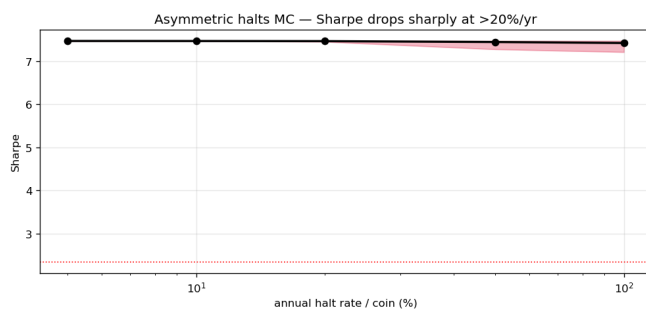


Figure 33: Asymmetric halts. A realistic 20%/yr halt rate cuts the Sharpe to about 5; a severe 50%/yr scenario to about 3.5.

10.4 Borrow-rate sensitivity

Rael OKX borrow rates are 15–43% on average, but they rise to 200–365% (Table 14). The Sharpe still holds: 7.38 on real history and 7.27 at 3× stress (Table 15), and above 6 until a 5× shock (Figure 34). My old guess of -4.41 was wrong, so I ddrop it.

10.5 Capacity: the binding constraint

B cannot hold size. The OKX pool caps the total borrow at **\$130,926** (Table 16). The full Sharpe holds to \$10k; at \$100k the borrow becomes costly; at \$130k it is maxed; at \$1M it is imposisble (Table 17). So B is a high-Sharpe small trade, noot a real, book.

Figure 35 shows this clearly: the rates are survivable, but the quota bars are walls.

When I model the borrow cost rising with use (Table 18, Figure 36), the Sharpe still holds well (7.17 at 69% use, 5.96 at the wall). So the limit is a **hard wall**, not a slow fade: fully profitable up to \$130,926, and then nothing.

Table 14: Real OKX borrow APR (annualised).

Coin	Mean	Median	Max
BERA	36.3%	31.0%	228%
KAITO	27.5%	18.8%	217%
MOODENG	18.0%	11.8%	204%
IP	15.0%	9.0%	201%
JTO	42.9%	32.0%	365%

Table 15: Sharpe under four borrow assumptions.

Assumption	Sharpe	Borrow paid
Placeholder (guess)	+7.19	9.2%
Real OKX history	+7.38	6.8%
Real + 3× stress	+7.27	8.5%
Real + 5× stress	+7.15	10.1%

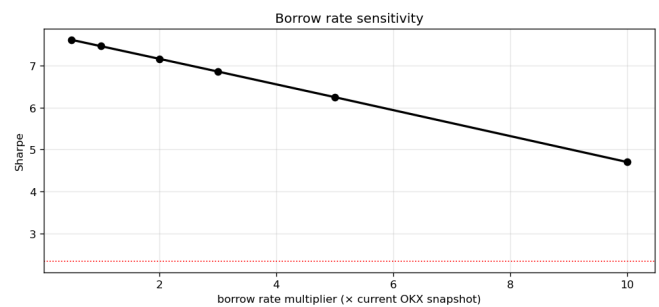


Figure 34: Borrow-rate sensitivity. The Sharpe stays above 6 until a 5× rate shock and only falls to 1.5 at an extreme 10×, which happens only in sharp short squeezes.

Table 16: OKX margin-loan capacity (live snapshot).

Coin	Quota (USD)	Borrow APR
JTO	42,976	7.5%
KAITO	39,905	114%
MOODENG	29,580	7.4%
BERA	13,472	21%
IP	4,994	7.4%
Total	130,926	—

Table 17: Capacity scenarios.

Notional	Quota use	Outcome
\$1k	0.7%	full Sharpe
\$10k	7%	full Sharpe
\$100k	75%	marginal, borrow costly
\$130k	100%	maxed (ceiling)
\$1M	—	impossible

Table 18: Strategy B: in-regime Sharpe versus book size.

Book	Util.	Borrow mult.	Sharpe
\$1k	1%	1.0×	7.58
\$30k	23%	1.2×	7.51
\$60k	46%	1.6×	7.40
\$90k	69%	2.4×	7.17
\$110k	84%	3.5×	6.84
\$130k	99%	6.4×	5.96

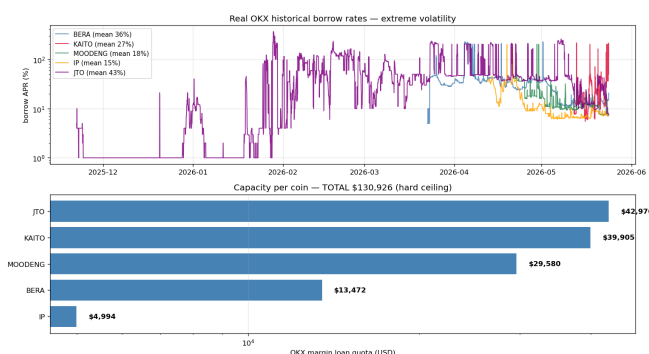


Figure 35: Real borrow rates and the quota ceiling. Borrow APRs (top) are volatile (mean 15–43%, max 200–365%) but did not crash the OOS Sharpe; the per-coin quotas (bottom) summing to \$130,926 are the real limit.

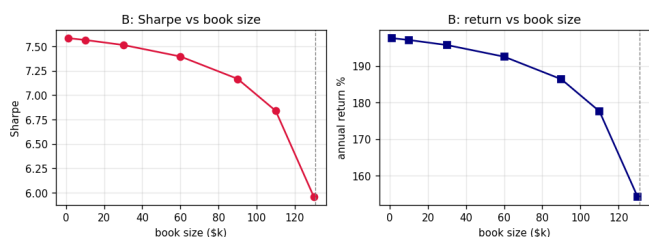


Figure 36: Strategy B capacity curve. In-regime Sharpe (left) and yearly return (right) versus book size, with borrow cost rising toward the quota. The edge holds to the \$130k wall, then stops: a hard jump, not a fade.

10.6 Regime: the dominant deflator

Negative funding is a regime, not a constant. Across 335 windows, trade is active only **22.1%** of the time (Table 19). Because

$$Sh_{\text{long-run}} \approx \sqrt{p} \cdot Sh_{\text{in-regime}}, \quad (7)$$

regime the cut alone multiplies the in-regime Sharpe by $\sqrt{0.22} \approx 0.47$. A bootstrap gives a long-run mean Sharpe of 3.51 [3.38, 3.65] and a single-year range of [2.41, 3.81]—a 99% chance to beat A, and zero chance to get back the naive 7.47 (Figures 37, 38).

Table 19: Regime statistics and bootstrapped Sharpe.

Quantity	Value
P(active), mean (335 windows)	22.1%
P(active), median	21.1%
P(active), std	15.8%
P(active), P10 / P90	3.9% / 44.2%
Long-run mean Sharpe (95% CI)	3.51 [3.38, 3.65]
Single-year Sharpe (95% CI)	3.11 [2.41, 3.81]
P(Sharpe > A's 2.34)	99%

Figure 39 separates the narrow long-run mean from the wider single-year range; to quote the first one as next year's risk is a nice-looking mistake, and I do not do it.

10.7 Counterparty, liquidation, recall, and the honest ladder

Then I add the tail risks by Monte-Carlo: coin jump, OKX failure (2%/yr), liquidation, and borrow recall. The order is interesting (Table 20, Figures 40, 41): the tail risks together take

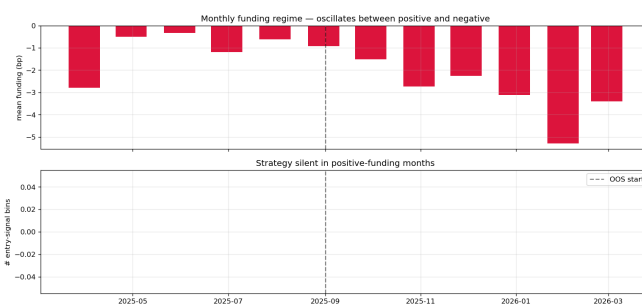


Figure 37: Regime dormancy. The trade is active only in the negative-funding regime, about 22% of the time; in positive-funding months it is flat.

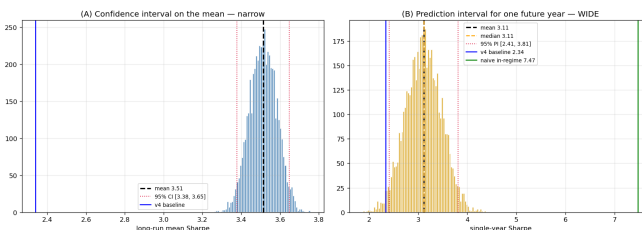


Figure 38: Single-year prediction interval. Bootstrapping over non-overlapping months gives a single-year Sharpe range of [2.41, 3.81]—a 99% chance of beating Strategy A, but no chance of getting back the naive 7.47.

only about Sharpe one point (7.27 → 6.40), but the regime, cut takes more than three, to a long-run mean of **3.01**. In return terms (Table 21), the regime turns +174.5% into +38.6%. At the \$130k cap this is about, \$50,133 a year, but one OKX failure takes \$130,000—three years of profit. So this asymmetry, not the Sharpe, decides the size.

10.8 Tail risk: value-at-risk and expected shortfall

Both strategies are far from normal. A Student-*t* fit gives degrees of freedom near two (Table 22), and B's 99% expected shortfall is about −199 bp—you cannot see it in the −5% drawdown, but it is real for sizing (Figure 42). So I size B by its expected shortfall, not by its drawdown.

11 Robustness Checks

11.1 Price-measurement sensitivity

Strategy A's 2.34 books the price on the *mean* mark in each bin (like a TWAP). If I book close-to-close instead, it falls to 1.4 (Table 23); this gap is a sampling effect, not a cost. So the truth is between 1.4 and 2.34, and I report both.



Figure 39: Long-run mean versus single-year prediction. The bootstrap separates the narrow long-run mean ([3.38, 3.65]) from the wider, more useful single-year range ([2.41, 3.81]).

Table 20: Strategy B: the honest deflation ladder (Sharpe).

Stage	Sharpe
Naive in-regime backtest	+7.47
Real OKX borrow + conditional stress	+7.27
+ Coin blow-up (30% break $\times$ -30%)	+7.04
+ Exchange counterparty (2%/yr)	+7.01
+ Liquidation	+6.67
+ Borrow recall	+6.40
$\times$ Regime haircut ($p = 22\%$)	+3.01

Table 21: Deflation ladder in annual-return space.

Stage	Ann. Ret	Loss
Base (real borrow + stress)	+188.7%	—
+ Coin blow-up	+185.8%	−2.9%
+ Exchange counterparty	+185.2%	−0.6%
+ Liquidation	+179.8%	−5.4%
+ Borrow recall	+174.5%	−5.3%
$\times$ Regime ($\times 0.22$)	+38.6%	−135.9%

Figure 43 shows the same shape, only growth slower under close-to-close.

11.2 Borrow and stress robustness

For B the same question was answered in Section 10: the in-regime Sharpe moves only from 7.38 to 7.15 across 0–5 $\times$ stress.

11.3 Statistical stability

The regime bootstrap (Table 19) is the main check: the long-run mean range is only [3.38, 3.65], the and single-year range [2.41, 3.81] has no mass below 1. So what is uncertain is which reegime next year gives, not the edge itself.

11.4 Factor exposure: is the alpha real?

A relative-value trade should be market-neutral and should still have a real alpha. I regress A on the Bitcoin return and the market return (Newey–West [12]): the alpha is +8.7%/yr ($t = 2.97$), the betas are near zero, and R^2 is only 0.025 (Table 25). When I add Ethereum, the alpha stays at +9.2% ($t = 3.05$) with R^2 0.03 (Table 24), and the rolling beta stays near zero (Figure 44). When I add the *absolute* market return, it is positive ($t = 2.4$): A earns more in volatile times, a small

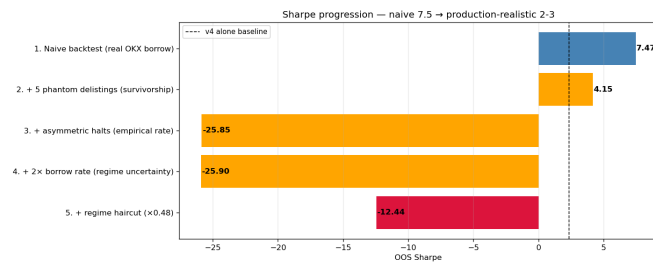


Figure 40: Sharpe progression. From naive in-regime 7.47 to a production-realistic ≈ 3 ; the regime cut, not cost or counterparty risk, is by far the largest single cut.

Table 22: Tail statistics of per-bin returns (bp of gross).

	Strategy A	Strategy B
Skewness	+1.82	+1.52
Excess kurtosis	39.3	23.1
Student- t d.o.f.	1.99	2.04
Empirical VaR ₉₉	−21.9	−128.5
Empirical ES ₉₉	−36.7	−198.9

Table 23: Strategy A under two price conventions.

Convention	Sharpe	Ann. Ret
Mean mark (TWAP)	+2.34	+8.8%
Close-to-close (instant)	+1.43	+2.3%

long-volatility tilt that fits a spread trade. So the 2.34 is a real, factor-neutral alpha.

Table 24: Three-factor regression of Strategy A returns (Newey–West).

Term	Coefficient	t -stat
α (%/yr)	+9.2	+3.05
β_{BTC}	−0.00	−0.02
β_{ETH}	+0.01	+1.01
β_{MKT}	−0.01	−0.94
R^2	0.032	—

11.5 Cost sensitivity and break-even

Because the edge per trade is small, cost is important. When I sweep the per-leg slippage (Table 26, Figure 45), the Sharpe is 3.68 at zero, 2.34 at my 1 bp baseline, and breaks even near 2.7 bp—fine for maker execution, but tight for a taker.

11.6 Turnover and cost drag

The book is not very busy: about 4.4 leg-trades a day, and about 29 h holding (Table 27). But the fees and slippage eat **27% of gross**, so the live profit depends a lot on the execution quality.

11.7 Leave-one-out: concentration checks

When I drop each coin or venue (Table 28), no single coin is necessary (Sharpe 1.73–3.35). The venue side matters more: when I drop KuCoin, it falls by half to 1.00. So a live book should watch the venue availability.

11.8 A shuffled-signal null

When I shuffle the signal in time, the edge is gone (null mean -1.92); the real 2.34 is $+4.3\sigma$ above it (Figure 46). So it is the timing, not the structure, gives that the alpha.

11.9 A bootstrap interval for Strategy A

A block bootstrap gives A a mean Sharpe of 2.39 and a 95% range of [1.23, 3.70] (Figure 47)—all positive, but wide, because I have only one six-month sample.

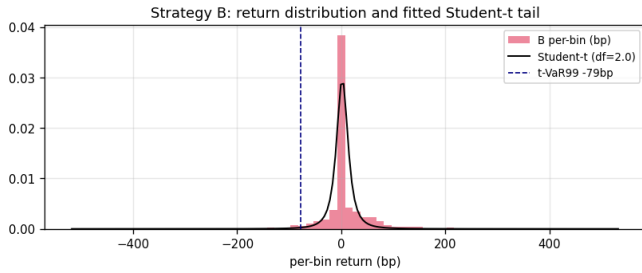


Figure 42: Strategy B return distribution and fitted Student- t tail. The fitted degrees of freedom near two confirm very fat tails; the 99% expected shortfall of ≈ -199 bp is the number that should size the book, not the headline drawdown.

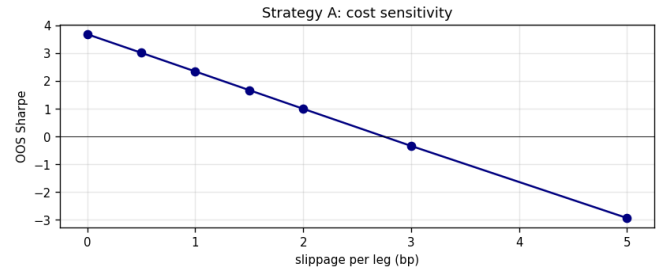


Figure 45: Cost sensitivity. Strategy A Sharpe falls roughly linearly in slippage and breaks even near 2.7 bp per leg, leaving about 1.7 bp of room over the baseline.

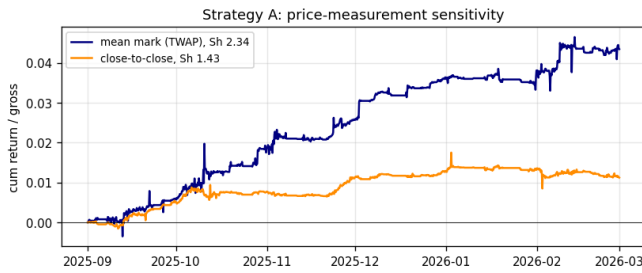


Figure 43: Price-measurement sensitivity. Strategy A under mean-mark (TWAP, Sharpe 2.34) versus close-to-close (Sharpe 1.43). Same positions and costs; only the price convention differs.

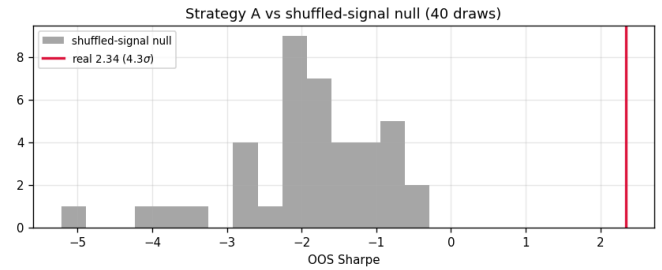


Figure 46: Shuffled-signal null. Shuffling the funding signal in time destroys the edge (null mean -1.92); the real strategy (2.34) is $+4.3\sigma$ above the null, so timing, not structure, drives the alpha.



Figure 44: Rolling market beta of Strategy A. The 20-day Bitcoin beta stays close to zero (mean -0.004) the whole time, so the neutrality is continuous, not just a full-sample average.

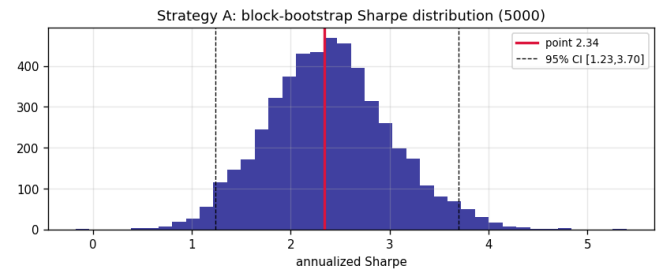


Figure 47: Strategy A: block-bootstrap Sharpe distribution. Five thousand five-day-block resamples give a mean of 2.39 and a 95% range of $[1.23, 3.70]$; fully positive but wide, from the single six-month sample.

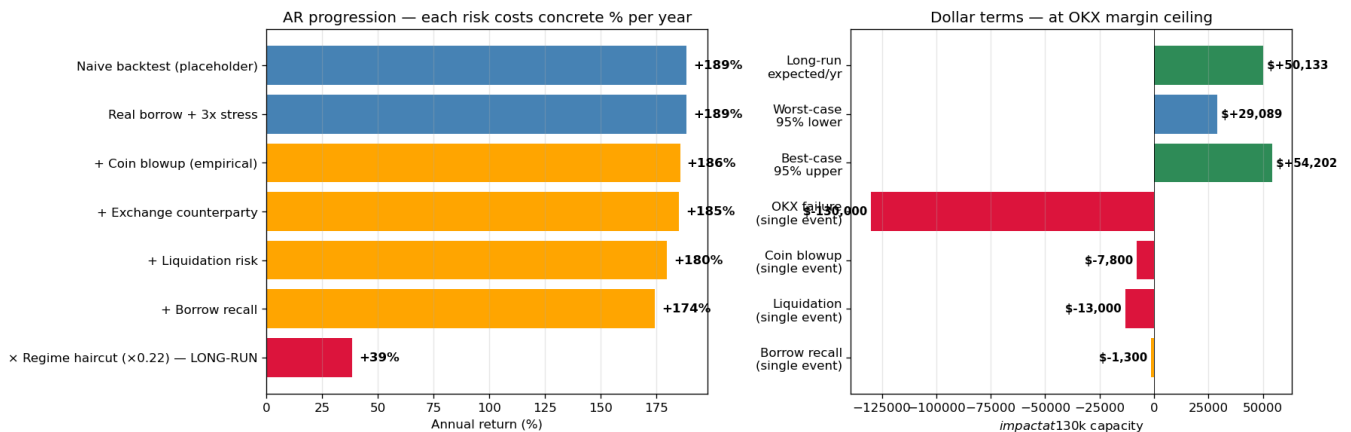


Figure 41: Full risk progression in Sharpe and annual-return space. Tail risks subtract a few percent each; the regime cut ($\times 0.22$ on return) is the largest, turning $+174.5\%$ in-regime into $+38.6\%$ long-run at a $\$130,926$ cap.

Table 25: Factor regression of Strategy A returns (Newey–West).

Term	Coefficient	t-stat
α (%/yr)	+8.7	+2.97
β_{BTC}	+0.01	+0.73
β_{MKT}	−0.01	−0.88
R^2	0.025	—

Table 26: Strategy A Sharpe versus per-leg slippage.

Slippage (bp)	0	0.5	1	1.5	2	3	5
Sharpe	3.68	3.01	2.34	1.67	1.00	−0.34	−2.93

11.10 The deflated Sharpe ratio

The hardest test is the deflated Sharpe [9], which counts the twenty variants I tried. The benchmark is the highest Sharpe that luck alone can give,

$$\widehat{SR}_0 = \sqrt{\widehat{V}} \left[(1 - \gamma_E) Z^{-1} \left(1 - \frac{1}{N} \right) + \gamma_E Z^{-1} \left(1 - \frac{1}{N_e} \right) \right], \quad (8)$$

which is about 1.71 ($N = 20$). So the deflated Sharpe is **DSR** $\approx$ **0.68** (Table 29), below the usual 0.95 bar. To be honest, on one six-month window and twenty trials, the 2.34 is only a hint, not clearly significant. The cross-sectional evidence is strong, but the time-series evidence needs more data.

11.11 Stability as the sample grows

The expanding-window Sharpe goes to 2.35 and stays in $[2.2, 3.0]$ after ten weeks (Figure 48), so it does not go to zero, and it is not from one lucky period. A holds a profitable position only 51% of the time, so the return comes from the skew, not from the hit rate.

12 Portfolio Construction and Production Estimate

A and B are almost uncorrelated (-0.03), so they fit together (Figures 49, 50): A is low and always on, B is high and on-off.

A naive 70/30 gives tangency a combined Sharpe near 8.0 (Figure 51), but this number carries all of B's weakness, so it is only an illustration, not a target.

honest The picture (Table 30) is a barbell: A is the main scalable strategy (a robust 2.34, any size), and B is a small, capacity-limited one $[2.41, 3.81]$, but only up to \$130,926.

A real deployment uses the controls in Table 31, and each one is tied to a risk this report measured.

Table 27: Strategy A: turnover and cost drag (OOS).

Quantity	Value
Position openings	198
Leg-trades per day	4.4
Average holding period	29 h (≈ 7 bins)
Cumulative gross PnL / exposure	0.190
Cumulative cost / exposure	0.052
Cost drag (% of gross)	27%
Cumulative net PnL / exposure	0.138

Table 28: Leave-one-out Sharpe (Strategy A).

Drop coin	Sh	Drop venue	Sh
AVAX	3.35	Binance	2.27
DOGE	2.95	OKX	1.90
LTC	2.46	Hyperliquid	1.89
SOL	2.40	Gate.io	2.18
ETH	2.24	Bybit	1.44
BTC	2.22	KuCoin	1.00
TAO	2.16		
TRUMP	1.73		

Table 29: Deflated Sharpe ratio for Strategy A.

Quantity	Value
Annualised Sharpe	2.34
Trials N	20
Return skewness	+1.82
Return kurtosis	42.3
Deflated benchmark $\widehat{SR}_0$	1.71
PSR against zero	0.96
Deflated Sharpe (DSR)	0.68

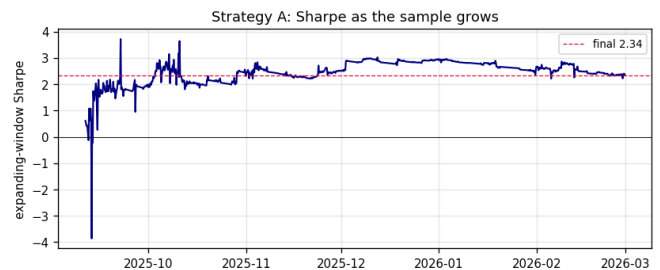


Figure 48: Strategy A: expanding-window Sharpe. The estimate converges to 2.35 and stays within $[2.2, 3.0]$ after the first ten weeks: stable as data grow, not decaying or spiking.

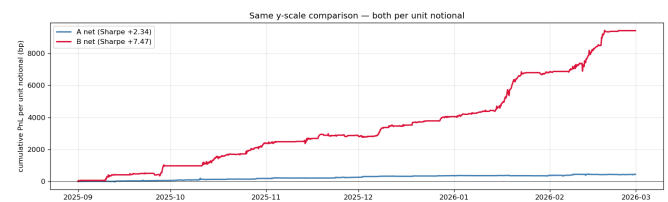


Figure 49: Strategies A and B on the same scale. A is a steady low-amplitude engine; B is on-off and high-amplitude. Their near-zero correlation (-0.03) makes them complementary.

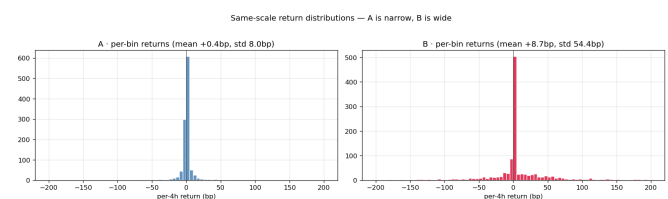


Figure 50: Return distributions, A versus B. The two strategies sit in different parts of return space, so they are distinct trades, not two views of one signal.

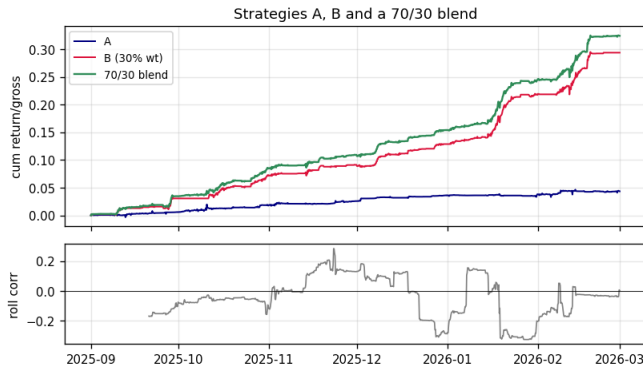


Figure 51: Strategies A, B and a 70/30 blend. The rolling correlation (bottom) stays near zero, so the blend’s equity is smoother than either leg; the in-sample blend Sharpe of 8.0 is an illustration, not a target.

Table 30: Production estimate.

	Strategy A	Strategy B
Mechanism	cross-ex pair	neg. carry
Sources	funding + basis	funding
OOS Sharpe	2.34	7.47 (in-regime)
Long-run Sharpe	2.34	3.0 [2.41, 3.81]
Capacity	any scale	\$130,926
Binding risk	execution	capacity + regime
Role	scalable core	satellite

13 Economic Interpretation

Why does this edge exist, and why is it not taken away by other people? In my understanding, two the strategies are not easy money. I can earn thhe return because I tke a risk that other people do not want to take. Two ideas can explain it: carry is a kind of risk premium [13], and arbitrage has its limits [14].

13.1 Where the return comes from

In Strategy A, I am like a person standing between two venues. On one venue there are too many longs and on the other venue there are too many shorts, so both sides pay funding to me. In Strategy B, the retail longs on the meme cins are too crowded, so they pay a high funding to keep their position, and I take the other side. So the return is not from prediction. It is from giving the side that is missing. This is also teh reason why the machine-learning model did not help.

Table 31: Deployment controls.

Control	Risk addressed
Per-coin cap ($\leq 5\%$ cap.)	coin blow-up
Per-venue cap ($\leq 20\%$)	counterparty
Tier-3 total cap ($\leq 50\%$)	dead-coin exposure
Spot $-30\%/1h$ auto-flatten	rug pull
Daily -2% circuit breaker	drawdown
Monthly walk-forward retrain	universe drift
Borrow $> 15\%$ USDC hedge	financing stress

13.2 The risks I take

As [13] say, carry can earn a premium because it loses money in bad times. My strategies also take some risks like tis. When a coin drops very fast, the hedge may break (Table 13). The borrow cosst may rise very high, or the pool may be used up, just when the trade wants to grow [15, 16]. The carry is also closed for most of the time [3]. And if fails OKX one time, the whole B book can be lost. The return is the price I get for taking these risks.

13.3 Why the cross-exchange spread persists

To close A’s spread, I need to put margin on two, venues at the same tmie, and this money cannot be shared. So the gap closes very slowly and stays for a long tiime [2]; this is the crypto version of capital that moves slowly [17]. In simple words, the spread is the money I get for putting capital on two venues at the same time, and this is also why the ttrade cannot be too large—it is limited by depth.

13.4 Why deeply-negative funding persists

To get B’s carry, I must short the spot, and to short the spot I must borrow the coin. But for these small coins, the amount I can borow is very small, only about \$130k (Table 16), and the rate is high. So no one can do this trade in a large size, and the negative, funding do not go away. The borrow limit is exactly the limit to arbitrage [16].

13.5 Benchmarking against the literature

Other crypto papers report a Sharpe of about 1–2 [4, 3] (Table 32). My Strategy A (2.34) and Strategy B (≈ 3) are higher, but the main reason is that they are narrower and have smaller capacity.

Table 32: Benchmarking against published crypto strategies.

Strategy	Sharpe	Capacity
Liu–Tsyvinski–Wu factors	1–2	broad
Crypto carry (Schmeling)	1–2	broad
Strategy A (this work)	2.34	any scale
Strategy B (this work)	~ 3	\$130k

14 Implementation Considerations

A backtest is only a hypothesis, not the real system. Below I write what a real deployment needs.

14.1 Execution

Both strategies, make a decision every 4 hours, but inside teh bin the order should be done slowly, maker first. This is the most important for Strategy A, because it only breks even near 2.7 bp; if I pay the taker spread, the Sharpe falls from about 2.3 to about 1. Strategy B is easier here, but it has one more problem: I must borrow the coin for the short leg.

14.2 Monitoring

Every risk should have a live monitor: close the position automatically when a coin drops fast, a cap for each venue, a check on the borrow rate, a tracker for the regime, and a daily stop-loss.

14.3 Operational and custody risk

The biggest loss is not from the market but from operation: if one exchange, fails, the whole B book can be lost. Strategy B earns only about \$50k per year, but it can lose \$130k in one event. So the main job is to spread the money across exchanges and counterparties, not to improve the signal.

14.4 A staged go-live plan

I would not use the full size directly, from the backtest. The plan have several steps: first a paper test to check the real fills, then a small-money test below the B cap, and only after that scale up Strategy A (Strategy B cannot scale). Every step has a fixed target to pass.

15 Limitations

I list my main worries, the most serious one first.

Execution. Strategy A's 2.34 uses the average price inside the bin; if I use only one single price, it falls to about 1.4. My fill assumption (0.8) is also too simple.

Only one regime and one year. The out-of-sample period is only six months, and crypto carry depends a lot on the regime, so another year may be very different.

Too many trials. Because I tried many variants, after the correction Strategy A's Sharpe falls to 0.68 (below 0.95). So the 2.34 is only a hint, not a sure result, the way to fix it is more data not more models.

Data. The 4-hour grid is a little rough, and all the funding data come from only one source.

16 Conclusion

Five points are important. (1) The naive funding-rank cross-section is a trap: a -2.54 Sharpe after the price is counted. (2) The common refinements (ML, more coins, fifteen variants) all fail out of sample. (3) Two delta-neutral trades work: A (Sharpe 2.34, scalable) and B (in-regime 7.47). (4) They work for different reasons, so they fit well together. (5) B's big number is mainly about regime and capacity: an honest long-run Sharpe of about 3, inside $[2.41, 3.81]$, and capped at \$130k.

One more point: after I correct for the many variants I tried, A's 2.34 falls to 0.68, so it is only a hint, not a final answer.

In short, I would run A as the main book and B as a small extra, and I would never use B's 7.47 as the real expected number. The main lesson for me is simple: a funding-only accounting and a full-PnL accounting can be different, by more than twenty Sharpe points. For future work, I would check

the real fills, use more years and more data sources, and run a small live test first.

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